Automated End-to-End 3-Axis CNC Machining: Task-Level Multi-Tool Planning, Differentiable Cutting Simulation, and Trajectory-Policy Co-Optimization

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Abstract—Advances in fused-deposition 3D printing have made additive fabrication accessible to non-experts: a user drops a digital mesh into slicer software, which automatically plans and executes trajectories in $\mathbb{R}^4$ $(x, y, z, \text{extrude})$. That accessibility has not transferred to 3-axis Computer Numerical Control (CNC) milling. Although 3-axis CNC trajectories inhabit a similarly compact continuous action space, subtractive machining requires first solving a combinatorial upstream task-planning problem—selecting raw stock material and dimensions, workholding orientation and clamping, multi-class cutting/drilling/threading tool libraries, feature-bound toolpath topologies, and narrow physically admissible bands of feeds and speeds—followed by long-horizon trajectory optimization through discontinuous Boolean material removal. Today, this hierarchy requires hours of manual configuration by expert CNC programmers. We challenge this paradigm with DiffCAM, an MIT-licensed, end-to-end automated 3-axis CNC machining framework that couples (1) task-and-motion-planning (TAMP) setup, workholding, and multi-tool operation selection over a 646-record clean-room tool catalog and physics-based feeds_and_speeds solver, (2) classical clean-room CAM trajectory initialization across facing, pocketing, trochoidal adaptive clearing, contouring, and peck drilling, and (3) dual-mode differentiable cutting simulation combining 3D swept-cylinder signed distance fields (SDFs) with Straight-Through Estimator (STE) stiffness continuation and chronological 2.5D cutter-workpiece engagement (CWE) radial chip-thinning mechanics. Across 16 single-tool benchmark tasks ($B = 64$ simulator calls), smooth-to-hard continuation cuts cycle time by $\mathbf{42.0 \pm 13.5}\%$ at 100% hard feasibility (4× evolutionary search), while first-order DiffPolicy-SHAC doubles the sample efficiency of zero-order Hard-PPO (**37.0**% vs. **18.5**%). Under strict matched-speed and optimal-chipload CWE evaluation (Exp5), DiffCAM-Chipload saves up to 7.9% cycle time at identical cutting feedrate via stock-verified air-detour shortcutting and **14.9**% overall while maintaining **97.7**% of engaged cutting inside the optimal undeformed chip-thickness band (h_{ex} RMSE 0.0053 mm). Furthermore, warm-starting from zero-shot SHAC (3.5 ms) or clean-room CAM seeds and rapidly overfitting directly to the test inference trajectory for $K \in \{8, \ldots, 32\}$ analytical CWE steps (Exp5b) achieves part-specific optimality in sub-second wall time (0.48–1.04 s, 95.6–98.5% optimal chipload)—a regime unreachable by test-time RL under comparable budgets. Finally, our automated multi-tool TAMP pipeline (Exp6) maps raw watertight .stl parts across three alloys directly into verified 4-tool Haas Next-Gen Control programs (O02001–O02004, up to 36.7% total / 43.7% cutting-time reduction) with 0.00 mm^3 gouge, alongside 48/48 six-dialect post-processor and CAMotics v1.2.0 watertight STL certification.

I. Introduction

Roboticists in the past have often dismissed the robotic challenges of manufacturing because the factory domain appears relatively well-defined and structured rather than addressing open-ended, general-purpose manipulation tasks. Yet manifesting physical objects from digital designs via machine tools has historically required deep artistic and technical expertise. Advances in desktop 3D printing have demonstrated that technical expertise in manufacturing processes is not inherently required to realize digital objects in the physical world: a user no longer needs to understand toolpaths, kinematics, or robotics, and can instead simply drop a digital model into a 3D printer's slicer software, which automatically plans and executes the trajectories required to extrude plastic in the shape of the part.

However, the plug-and-play accessibility of 3D printing software has not transferred to more complex and capable manufacturing processes, such as 3-axis CNC milling machines. At the continuous motion level, the instantaneous action space for a typical fused-deposition modeling (FDM) 3D printer is $\mathbb{R}^4$ $(x, y, z, \text{extrude})$, and the trajectory action space for 3-axis CNC machining is superficially similar—$\mathbb{R}^4$ $(x, y, z, \text{spindle/cutting speed})$. Relative to 3D printing, CNC machining allows the fast and repeatable creation of high-strength functional parts out of both common and exotic structural materials. However, because machining is a subtractive process—in which a rotating cutter permanently carves away solid metal and every motion irreversibly alters future cutting loads [1, 20]—vastly more upstream planning is required prior to choosing a continuous trajectory:

1) Stock and Workholding: One must first select a piece of raw material, called stock (including alloy composition and billet dimensions), choose the orientation of the stock inside the CNC work envelope, and determine how it is rigidly clamped down against cutting forces, called workholding (e.g., parallel vise jaws that themselves become collision obstacles).
2) Multi-Class Tool Libraries and Operation Topologies: Whereas FDM 3D printers typically operate

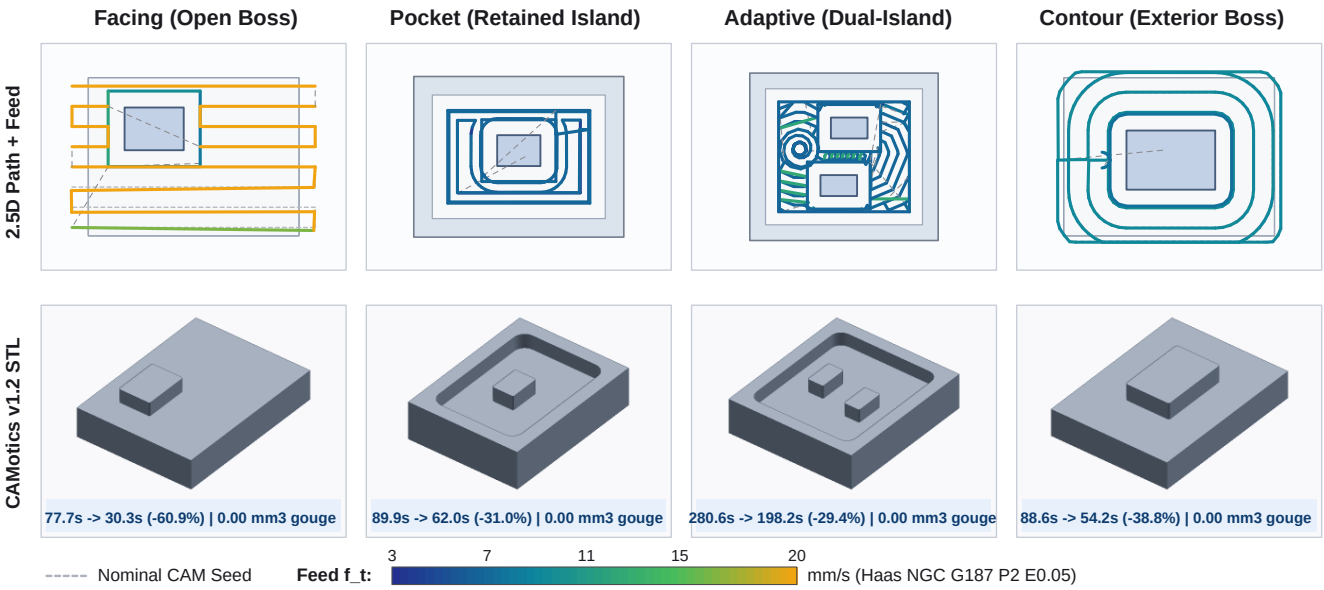


Fig. 1. DiffCAM joint 2.5D toolpath and feedrate co-optimization certified in CAMotics v1.2.0. Top row: Representative held-out test parts across four modular MIT clean-room CAM strategy families showing stock billets, retained islands/bosses, dashed gray nominal CAM seeds (37–370 segments), and optimized STE-Continuation-Adam trajectories colored by per-segment feedrate $f_t \in [2.8, 20.0]$ mm/s under Haas Next-Gen Control corner blending (G54, G187 P2 E0.05). Bottom row: 4×-MSAA isometric renderings of the closed, watertight binary STL meshes reconstructed directly by containerized CAMotics v1.2.0 (0.25 mm grid) from posted Haas NGC G-code, achieving 29.4%–60.9% cycle-time reductions on these four parts (39.5% mean across all 8 test parts) with 0.00 mm^3 target gouge, 0.00 mm^3 residual, and exact STL volume identity with nominal CAM.

with a single nozzle diameter and one filament corresponding to a single primitive operation (extruding plastic), cost- and time-efficiently removing material from metal stock requires a diverse library of tools spanning distinct classes of operations: cutting tools (flat, bull-nose, and ball end mills, face mills), hole-making tools (spot drills, twist drills, reamers), and threading tools (taps, thread mills). Each tool is paired with specialized parameterized toolpaths: for instance, drills can plunge through shallow stock in a single motion or execute small incremental cutting motions and periodically retract to evacuate chips before plunging deeper (peck drilling); cutting tools can slice a thin datum layer off the top of the stock via a facing toolpath, hog out bulk material rapidly at lower precision via a roughing or trochoidal toolpath [8, 14], or clean tight concave corners left behind by larger tools via rest-machining.

3) Narrow Admissible Feeds and Speeds: Every combination of workpiece alloy, cutter substrate, flute geometry, and local radial/axial engagement admits only a narrow physical operating window for the rotational speed of the spindle relative to linear motion against the workpiece, known as feeds and speeds [1, 12]. Feeding too slowly below minimum chip thickness causes rubbing and work-hardening; feeding too aggressively exceeds chip-evacuation, spindle torque, or cantilever tool-deflection limits and snaps the cutter.

The combination of stock material, workholding, tool selection, toolpath topology, and feeds and speeds creates a massive hierarchical planning space that must be resolved before optimizing any continuous trajectory. Today, people solve the complexity of CNC machining in an overwhelmingly manual way: expert CNC programmers spend many hours in visual computer-aided manufacturing (CAM) interfaces manually configuring stock billets, fixture clearances, tool-library assemblies, feature-associated parameterized toolpaths, and handbook or calculator-derived feeds and speeds.

We challenge this manual paradigm and propose that 3-axis CNC machining should be solvable end-to-end in an automated way—at a quality level that currently demands hours of expert programmer time—using modern learning-based and gradient-based approaches that combine task-level setup planning prior to trajectory optimization with classical CAM trajectory initialization and differentiable cutting simulation. Realizing this vision requires overcoming three fundamental technical barriers at the physics-and-learning interface:

1) Discontinuous Boolean Material Removal and Surrogate Drift: Subtracting a swept cutter volume ψ_{tool} from an in-process workpiece (IPW) signed distance field ϕ_t via $\phi_{t+1} = \max(\phi_t, -\psi_{\text{tool}})$ [18, 21] yields identically zero spatial gradients in empty air and non-differentiable kinks at stock boundaries, while naive smooth relaxations suffer volumetric occupancy drift [13, 19] and relative waypoint deltas

accumulate $O(T)$ drift across $T \sim 37$–370 segments.

2) Controller- and Chipload-Blind Kinematics: Optimizing path length at constant speed ignores physical CNC corner blending (RS274 G64, Haas G187) [4, 11], whereas optimizing volumetric removal without radial chip-thinning mechanics [12] distorts actual undeformed chip thickness h_{ex} during partial-immersion cuts.
3) Per-Part Planning Latency vs. Generalized RL Suboptimality: Training an amortized RL policy yields millisecond zero-shot inference on unseen parts, whereas direct per-part differentiable optimization tailors every waypoint and feedrate to the exact part geometry—raising the question of whether fast test-time trajectory overfitting can unify both regimes.

To address both the upstream setup bottleneck and downstream trajectory physics, we present DiffCAM (Fig. 1), an MIT-licensed automated 3-axis machining stack. Our contributions are:

- Automated Multi-Tool Task-and-Motion Planning (TAMP): We build a 100% clean-room MIT catalog of 646 tools and holders across 15 library groups and 32 entity types (StandardToolLibrary), paired with a clean-room feeds_and_speeds physics engine and an .stl-driven TAMP planner (cam.tamp_planner) that automatically selects stock margins, vise-jaw workholding zones, multi-tool roughing/rest-finishing/spot-drilling/peck-drilling sequences, and persistent-IPW trajectories.
- Dual-Mode Differentiable Cutting Simulation (3D Swept-SDF & Chronological 2.5D CWE): We formulate both (a) a GPU-accelerated Taichi [9] 3D swept-CSG simulator with Straight-Through Estimator (STE) [2] stiffness continuation ($k : 25 \rightarrow 90$), acceleration-bounded corner blending, and volumetric Peak MRR barriers, and (b) a subsegmented chronological 2.5D dexel CWE optimizer (SubsegmentedCWEModel) modeling Martellotti radial chip thinning, slotting derating, cantilever deflection, and stock-verified empty-air shortcutting.
- Controlled RL Benchmark and Fast Inference-Time Trajectory Overfitting: Across 16 tasks and 8 methods under strict simulator-call parity, we quantify first-order (STE-Continuation-Adam, DiffPolicy-SHAC [22]) vs. zero-order (Hard-PPO [17], Hard-CEM [15], Hard-CMA-ES [7]) optimization, separate fair same-speed geometric gains from optimal-chipload gains (Exp5), and show that warm-starting from zero-shot SHAC followed by $K \in \{4, \ldots, 32\}$ analytical trajectory-overfitting steps (Exp5b) rapidly outperforms both zero-shot RL and test-time PPO fine-tuning.
- 6-Controller G-Code, CAMotics STL, and Live 3-Way WebGL Verification: Solutions are verified across 6 CNC dialects (48/48 pass), containerized CAMotics v1.2.0 [3] (8/8 watertight STLs, 0.00 mm^3 gouge), multi-tool Haas NGC packets (O02001–O02004), and an interactive 3-way side-by-side WebGL stock-carving simulator.

II. Related Work

Task and Motion Planning (TAMP) and Feature Recognition. Integrated task and motion planning couples discrete symbolic decisions—such as object, grasp, or tool selection—with continuous geometric feasibility and trajectory optimization [6, 10]. While TAMP has been widely studied in pick-and-place, assembly, and robotic extrusion, subtractive CNC process planning has largely remained split between rule-based CAD feature recognition and manual CAM setup [20]. DiffCAM brings a TAMP formulation directly to 3-axis subtractive milling: discrete search over a multi-class tool library balances bulk roughing MRR against physical automatic-tool-changer (ATC) swap latency while passing an evolving persistent 2.5D IPW heightfield across consecutive tools so downstream rest-finishing trajectories adapt continuously to upstream stock removal.

Geometric CAM, Engagement Control, and Feeds/Speeds. Classical pocket and contour milling rely on 2D Voronoi medial-axis offsets [8] and parallel raster passes, which suffer sharp radial immersion spikes in concave corners. Trochoidal and adaptive strategies [1, 14] bound engagement geometrically, while solid verification uses Z-buffer dexels [21], R-functions [18], or octree isosurfaces [3]. Independently, metal-cutting mechanics establishes that when radial width of cut a_e is below cutter radius $R = D/2$, the maximum undeformed chip thickness h_{ex} thins according to Martellotti's kinematic relation [1, 12], requiring elevated feed per tooth f_z to prevent rubbing while respecting spindle power and cantilever beam deflection limits. Prior feedrate schedulers adjust speed along a frozen path [4]; DiffCAM jointly optimizes continuous XY waypoints, stock-verified air shortcuts, and per-station feeds through differentiable CWE and corner-blending kinematics [11].

Differentiable Simulation, Amortized RL, and Test-Time Optimization. Differentiable physics engines [5, 9] enable low-variance first-order trajectory optimization and short-horizon analytic actor-critic (SHAC) policy learning [22]. However, discontinuous contact or Boolean clipping can induce gradient variance or surrogate bias [13, 19], motivating comparisons with zero-order PPO [16, 17], CEM [15], and CMA-ES [7]. Beyond offline training, our work highlights a key structural advantage of differentiable simulation over model-free RL at deployment: whereas fine-tuning an RL policy on a single test instance requires hundreds of noisy rollouts, warm-starting from an amortized neural policy and taking a handful of analytical simulator gradient steps directly on the inference trajectory eliminates policy approximation error in milliseconds.

III. Hierarchical Task Planning and Differentiable Cutting Mechanics

A. Task-Level Setup, Clean-Room Tool Library, and Persistent-IPW Planning

Given an input watertight 3D triangle mesh $\mathcal{M} \subset \mathbb{R}^3$ (.stl), workpiece alloy identifier m_{stock}, and machine envelope $\mathcal{E}_{\text{mach}}$, our upstream TAMP module (cam.tamp_planner) resolves setup and discrete multi-tool sequencing prior to continuous optimization:

1) Stock, Orientation, and Workholding Synthesis: Aligning primary feature accessibility with $+Z$, the planner selects billet bounds $[x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [-L_z, 0]$ incorporating top-facing datum stock δ_{top} (1.0–1.5 mm) and bottom parallel vise-jaw clamping grip δ_{grip} (2.0 mm), instantiating explicit forbidden vise-jaw collision boxes $\mathcal{V}_{-Y}, \mathcal{V}_{+Y}$ along the clamped faces.
2) 2.5D + Hole Geometric Feature Recognition: Rasterizing upward-facing facets ($n_z > 0$) onto a 0.25 mm G54 heightfield $z_{\text{tgt}}(x, y)$ extracts top-facing depth, exterior boss steps, cylindrical blind/through holes $\{(x_h, y_h, D_h, z_{\text{top}}, z_{\text{bot}})\}$ (identified via aspect-ratio and $\pi/4$ bounding-box circularity invariants), and enclosed 2.5D pockets/islands along with minimum channel width $w_{\min}$ and concave corner fillet radius $r_{\min}$.
3) Discrete Tool-Library Search with ATC Cost: Querying our clean-room StandardToolLibrary ($\mathcal{T}$, containing 574 metric/inch milling, drilling, and threading tools plus 72 BT30/CAT40/HSK holders across 15 library groups and 32 entity types), the planner bounds maximum finishing endmill diameter by $D_{\text{fin}} \leq \min(2r_{\min}, w_{\min} - \delta_c)$ and evaluates whether prepending a larger rigid roughing endmill $D_{\text{rough}} > D_{\text{fin}}$ reduces total machining time once physical automatic-tool-changer swap overhead $t_{\text{ATC}} = 6.0$ s is charged:

$$J_{\text{task}}(\mathcal{S}) = \sum_{o \in \mathcal{S}} \frac{\widehat{V}_o}{\text{MRR}(m_{\text{stock}}, \tau_o)} + (|\mathcal{U}(\mathcal{S})| - 1)\, t_{\text{ATC}}, \tag{1}$$

where $\mathcal{U}(\mathcal{S})$ is the set of unique tools in operation sequence $\mathcal{S}$ and $\text{MRR}(m_{\text{stock}}, \tau_o)$ is computed by our clean-room feeds_and_speeds physics solver. Top facing and exterior boss contouring reuse the selected primary endmill (T_1) to incur zero extra tool change, followed if beneficial by T_2 corner/island rest-finishing, T_3 90° spot-drill chamfering, and T_4 diameter-matched carbide twist drilling (automatically switching to G83 multi-peck chip-clearing cycles whenever hole aspect ratio $L_h/D_h > 2.2$).
4) Persistent In-Process Workpiece (IPW) Propagation: Rather than optimizing each operation on an idealized fresh block, the evolving 2.5D dexel stock field $z^{(o)}_{\text{IPW}}(x, y)$ carved by operation o is passed directly as the initial stock state $z^{(o+1)}_{\text{IPW}} \leftarrow z^{(o)}_{\text{final}}$ to operation $o{+}1$, enabling rest-finishing tools to bridge rapidly across already-evacuated pocket interiors and regulate chip load strictly inside uncut corner cusps.

B. Dual-Mode 3D Swept-Cylinder SDF and STE Continuation

For general 3D volumetric verification and first-order trajectory/RL benchmarking, we represent the stock billet on a uniform Eulerian grid $\Omega \subset \mathbb{R}^3$ of pitch Δx with signed distance values $\phi_t : \Omega \to \mathbb{R}$ after segment $t \in \{0, \ldots, T\}$ ($\phi_t(\mathbf{x}) < 0$ inside material). For a flat-end cutter of radius R, flute length L_{flute}, holder clearance L_{clear}, and holder radius $R_{\text{holder}} > R$ sweeping linearly from $\mathbf{p}_{t-1}$ to $\mathbf{p}_t$ ($\Delta\mathbf{p}_t = \mathbf{p}_t - \mathbf{p}_{t-1}$), let $\tau^\star(\mathbf{x}) = \text{clip}\big(\frac{(\mathbf{x}-\mathbf{p}_{t-1})\cdot\Delta\mathbf{p}_t}{\|\Delta\mathbf{p}_t\|_2^2+\varepsilon}, 0, 1\big)$ and $\mathbf{c}_t(\mathbf{x}) = \mathbf{p}_{t-1} + \tau^\star(\mathbf{x})\Delta\mathbf{p}_t$. The swept cutting-flute SDF is

$$\psi_{\text{tool}}(\mathbf{x}; \mathbf{p}_{t-1}, \mathbf{p}_t) = \max\Big(\|\mathbf{x}^{xy} - \mathbf{c}_t^{xy}(\mathbf{x})\|_2 - R, \; c_t^z(\mathbf{x}) - x^z, \; x^z - \big(c_t^z(\mathbf{x}) + L_{\text{flute}}\big)\Big). \tag{2}$$

Exact Boolean subtraction updates $\phi_t^{\text{hard}}(\mathbf{x}) = \max\big(\phi_{t-1}^{\text{hard}}(\mathbf{x}), -\psi_{\text{tool}}(\mathbf{x})\big)$. To obtain dense analytical derivatives across stock boundaries without suffering low-stiffness volumetric inflation, we pair exact Boolean evaluation with a numerically stabilized log-sum-exp smooth surrogate at stiffness $k > 0$:

$$\phi_t^{(k)}(\mathbf{x}) = m + \frac{1}{k}\log\Big(e^{k(\phi_{t-1}^{(k)}(\mathbf{x})-m)} + e^{k(-\psi_{\text{tool}}(\mathbf{x})-m)}\Big), \tag{3}$$

where $m = \max(\phi_{t-1}^{(k)}(\mathbf{x}), -\psi_{\text{tool}}(\mathbf{x}))$ and smooth occupancy is $\sigma_k(\phi) = (1 + e^{k\phi})^{-1}$. Across optimization step $n \in \{0, \ldots, N-1\}$, we anneal $k(n)$ linearly from $k_{\min} = 25$ to $k_{\max} = 90$ inside a Straight-Through Estimator (STE) [2]:

$$\mathcal{L}_{\text{STE}}(\boldsymbol{\theta}; k(n)) = \mathcal{L}_{\text{hard}}(\boldsymbol{\theta}) + \Big(\mathcal{L}_{k(n)}(\boldsymbol{\theta}) - \text{sg}\big[\mathcal{L}_{k(n)}(\boldsymbol{\theta})\big]\Big), \tag{4}$$

where $\text{sg}[\cdot]$ stops gradients. Forward model selection and feasibility checkpointing are thus governed strictly by exact Boolean $\mathcal{L}_{\text{hard}}$, while reverse-mode Taichi kernels propagate $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{k(n)}$.

C. Blended Acceleration-Bounded CNC Kinematics

Physical CNC lookahead interpolators avoid full stops (G61) at collinear or gentle G1 junctions by blending velocity vectors within a path tolerance (RS274 G64 P0.05, Haas G187 P2 E0.05) [4, 11]. Given segment chord $\ell_t = \|\Delta\mathbf{p}_t\|_2$, unit tangent $\hat{\mathbf{d}}_t$, commanded speed v_t (f_t on G1, v_{rapid} on G0), and velocity vector $\mathbf{v}_t = v_t\hat{\mathbf{d}}_t$, acceleration-bounded junction transition duration over machine limit $a_{\max}$ yields smooth, closed-form segment durations

$$\tau_t(\mathbf{p}, \mathbf{f}) = \frac{\sqrt{\ell_t^2 + \varepsilon_L^2} - \varepsilon_L}{v_t} + \frac{\sqrt{\|\mathbf{v}_t - \mathbf{v}_{t-1}\|_2^2 + \varepsilon_v^2} - \varepsilon_v}{2\, a_{\max}}, \tag{5}$$

plus terminal deceleration $\|\mathbf{v}_T\|_2/(2a_{\max})$, with total cycle time $T_{\text{cycle}} = \sum_{t=1}^{T} \tau_t(\mathbf{p}, \mathbf{f})$. Equation (5) is analytically differentiable with respect to both waypoint coordinates $\mathbf{p}$ and per-segment speeds $\mathbf{f}$.

D. Chronological 2.5D Cutter-Workpiece Engagement (CWE) and Chipload Physics

While constraining global volumetric removal $\text{MRR}_t = \Delta V_t/\tau_t \le \text{MRR}_{\max}$ prevents gross spindle overload on 3D roughing passes (Eq. (7)), evaluating trajectories under realistic metal-cutting physics reveals a subtle limitation of MRR-only surrogates: during light radial skimming or corner rest-milling ($a_e \ll D/2$), radial chip thinning drastically reduces the actual undeformed chip thickness h_{ex} cut by each flute [12].

To optimize directly for physical chip load and minimal cycle time, we formulate SubsegmentedCWEModel (algorithms.chipload_opt) coupled to our clean-room cam.feeds_and_speeds solver:

1) Material- and Tool-Specific Reference Physics: Given stock alloy properties (Brinell hardness, specific cutting energy k_p, carbide surface speed SFM_0, base chip load $f_{z,0}$, slotting factor) and cutter geometry (D, flute count z_f, helix angle β_h, flute length, stick-out L_{stick}, substrate Young's modulus E), our clean-room solver computes spindle speed N_{RPM}, baseline feed f_{base}, and target undeformed chip thickness h_{opt} while enforcing six simultaneous physical caps: machine $f_{\max}$, spindle power $P_k \le P_{\max}$, machine and tool-core yield torque, cutting force $F_{c,k} \le F_{\max}$, and cantilever endmill deflection $\delta_k = \frac{F_{c,k} L_{\text{stick}}^3}{3EI} \le \delta_{\max}$.
2) Chronological Sub-Segmented CWE and Radial Chip Thinning: Long cutting moves are subdivided to $\le$ 2.0 mm stations ($k = 1, \ldots, M$). Sweeping each station chronologically over the 0.25 mm 2.5D dexel heightfield z_{stock} yields exact removed volume ΔV_k, axial depth $a_{p,k}$, and effective radial immersion $u_k = \text{clip}(a_{e,k}/D, 0, 1)$. By Martellotti's kinematics [12], the radial chip-thinning ratio $\psi(u_k)$ and actual undeformed chip thickness $h_{\text{ex},k}$ at tooth-passing frequency $f_{\text{tooth}} = (N_{\text{RPM}}/60)\, z_f$ are
$$\psi(u_k) = \begin{cases} \sqrt{4u_k(1-u_k)}, & 0 < u_k < 0.5, \\ 1, & u_k \ge 0.5, \end{cases} \qquad h_{\text{ex},k} = \frac{f_k}{f_{\text{tooth}}}\, \psi(u_k). \tag{6}$$
In the forward pass, $\Delta V_k, a_{p,k}, u_k$ use exact hard dexel state; in the backward pass, STE analytical spatial sensitivities $\partial \Delta V_k/\partial \mathbf{p}$ propagate lateral engagement gradients so waypoints shift smoothly to equalize u_k while per-station feeds f_k track $h_{\text{ex},k} \to h_{\text{opt}}$ on engaged cuts and sprint at $f_{\max}$ across zero-engagement air segments.
3) Stock-Verified Empty-Air Detour Shortcutting: After continuous STE-Adam convergence, contiguous runs of horizontal G1 stations verified against z_{stock} to remove $\Delta V_k = 0$ are collapsed into collision-checked G0 clearance-plane bridges or straight G1 chords, saving cycle time even when cutting feedrates are locked identically across methods.

For 3D SDF benchmarking (Exp1–Exp4), the composite loss combines normalized cycle time, target gouge/residual, volumetric Peak MRR overload $\mathcal{B}_{\text{MRR}} = w_{\text{MRR}} \sum_{t=1}^{T} \max(0, \text{MRR}_t/\text{MRR}_{\max} - 1)^2$, exact 2D wall-clearance $\mathcal{B}_{\text{wall}}$, cylindrical holder-collision $\mathcal{B}_{\text{holder}}$, and rapid-stock $\mathcal{B}_{\text{rapid}}$ barriers:
$$\mathcal{L}_{\text{3D}} = w_T \frac{T_{\text{cycle}}}{T_{\text{nom}}} + w_g \bar{V}_{\text{gouge}}^2 + w_r \bar{V}_{\text{res}}^2 + \mathcal{B}_{\text{MRR}} + \mathcal{B}_{\text{wall}} + \mathcal{B}_{\text{holder}} + \mathcal{B}_{\text{rapid}}, \tag{7}$$
while CWE chipload co-optimization (Exp5–Exp6) minimizes $\mathcal{L}_{\text{CWE}} = \frac{T_{\text{cycle}}}{T_{\text{nom}}} + w_h \sum_{k \in \mathcal{E}} \frac{\tau_k}{T_{\text{nom}}} \left(\frac{h_{\text{ex},k} - h_{\text{eff},k}}{h_{\text{opt}}}\right)^2 + \mathcal{B}_{\text{MRR}} + \mathcal{B}_{\text{res}} + \mathcal{B}_{\text{smooth}}$.

IV. Per-Part Optimization, Generalized RL, and Fast Inference-Time Overfitting

A. Drift-Free Waypoint and Column-Group Co-Parameterizations

Perturbing relative displacements $\Delta \mathbf{p}_t$ induces $O(T)$ cumulative random-walk drift across 37–370 segments, driving downstream finishing passes into blueprint walls. We enforce zero-gouge kinematic structure by holding layer heights $\bar{p}_i^z$ fixed and bounding horizontal excursions by exact analytical Euclidean wall clearance $d_{\text{wall}}(\bar{\mathbf{p}}_i)$:
$$\mathbf{p}_i^{xy}(\boldsymbol{\theta}) = \bar{\mathbf{p}}_i^{xy} + r_i \tanh(\mathbf{u}_{xy,i}), \qquad r_i = \text{clip}\big(d_{\text{wall}}(\bar{\mathbf{p}}_i) - \delta_{\text{pin}},\, 0,\, r_{\max}\big). \tag{8}$$
In our 3D spline model (JointPathFeedSpline), $\mathbf{u}_{xy} = \mathbf{B}\boldsymbol{\theta}_{\text{pos}}$ uses a clamped cubic B-spline basis ($K_p \le 40$, $r_{\max} = 1.25$ mm) with plunge/retract pinning; in our sub-segmented CWE model (SubsegmentedCWEModel), interior waypoints sharing vertical plunges/retracts are coupled into G vertical-column groups ($r_{\max} = 3.0$ mm) so vertical plunges remain strictly vertical while open-air lead-ins retract freely, paired with bounded sigmoid cutting feeds $f_k \in [f_{\min}, 0.995\, v_{\max}]$.

B. Generalized Residual RL vs. Fast Inference-Time Trajectory Overfitting

A central architectural question for learning-based manufacturing is whether DiffCAM trains a generalized multi-part policy or optimizes directly on each target part—and how the two paradigms interact at deployment:

1) Direct Per-Part Differentiable Optimization (DiffCAM STE-Adam / DiffCAM-Chipload): No neural network weights are trained across parts; instead, given any new part's initialized trajectory $\bar{\boldsymbol{\tau}}$, STE-Adam optimizes $\boldsymbol{\theta} \in \mathbb{R}^D$ ($D = 2G + M \approx$ 250–650) directly against the differentiable simulator on that part (1–3 s CPU/GPU).

2) Amortized Generalized RL Policies (DiffPolicy-SHAC / Hard-PPO): A compact tanh MLP $\boldsymbol{\pi}_\psi(\mathbf{a}_t \mid \mathbf{s}_t)$ (18-D global SDF state or 12-D local CWE state) is trained offline across development parts using either first-order analytical simulator gradients (SHAC [22]) or zero-order clipped policy gradients with GAE (PPO [16, 17]). On an unseen test part $\mathcal{P}_{\text{test}}$, a single forward pass produces $\boldsymbol{\theta}_0 = \boldsymbol{\pi}_{\psi^\star}(\mathbf{S}(\mathcal{P}_{\text{test}}))$ in <2.5 ms, but suffers finite-capacity approximation error across varying pocket/channel geometries.
3) Hybrid Warm-Start Inference-Time Trajectory Overfitting (SHAC + DiffCAM-Overfit-K): Because our simulator is analytically differentiable at test time, we do not need to stop at $\boldsymbol{\theta}_0$ nor fine-tune ψ via noisy policy gradients. Instead, we initialize $\boldsymbol{\theta}^{(0)} \leftarrow \boldsymbol{\pi}_{\psi^\star}(\mathbf{S}(\mathcal{P}_{\text{test}}))$, detach the MLP bottleneck, and apply $K \in \{4, 8, 16, 32\}$ first-order CWE STE-Adam updates directly in trajectory space $\mathbb{R}^D$, followed by stock-verified air-detour shortcutting. Whereas zero-order RL fine-tuning on $\mathcal{P}_{\text{test}}$ faces gradient-estimator variance scaling as $\mathcal{O}(D/(\sigma^2 N_{\text{pop}}))$—rendering 16–160 rollouts ineffective—exact per-station partials $\partial\mathcal{L}_{\text{CWE}}/\partial f_k$ and $\partial\Delta V_k/\partial\mathbf{p}_g$ allow $K = 8$–16 steps (200–450 ms) to rapidly overfit to the exact unseen geometry and restore $> 96\%$ optimal chipload compliance.

V. Experimental Setup, Split Protocol, and Metrics

A. Benchmark Suites, Clean-Room Catalogs, and Simulators

Single-Tool 16-Task Suite (Exp1–Exp5b): Covers four clean-room 2.5D CAM strategy engines—facing_open_boss (37–39 segs), pocket_island (92–169 segs), adaptive_channel (261–370 segs), and contour_profile (156–188 segs)—with 2 dev and 2 held-out test geometries per family (16 tasks, Al 6061-T6 billets $30\text{–}38 \times 24\text{–}32 \times 8$ mm, $D = 5.0$ mm 3-flute coated-carbide flat endmill at 8000 RPM, $v_{\max} = 20$ mm/s, $v_{\text{rapid}} = 100$ mm/s, $a_{\max} = 150$ mm/s^2). Multi-Tool .stl TAMP Suite (Exp6): Four multi-feature 3D parts supplied as raw binary .stl solids across three alloys—tamp-flanged-bracket (Al 6061-T6), tamp-sensor-housing (Al 6061-T6, central island), tamp-manifold-plate (Al 7075-T6, $r_{\min} = 1.5$ mm tight corners + pocket-floor holes), and tamp-stepped-mount (Mild Steel 1018)—combining top facing, exterior steps, pockets, islands, spot chamfering, and G83 peck drilling.

B. Role of dev vs. Held-Out test Splits

To ensure strict methodological clarity across direct trajectory optimization and policy learning:

- Direct Per-Part Trajectory Optimization (STE-Continuation-Adam, Annealed-Adam, DiffCAM-Chipload, Hard-CEM/CMA-ES): Uses the 8 dev tasks solely to select and freeze solver hyperparameters (learning rates, barrier weights, k-schedule). On the 8 held-out test tasks, each optimizer starts cold from that test part's clean-room CAM seed without seeing any test geometry during tuning.
- Generalized RL Policies (DiffPolicy-SHAC, Hard-PPO, SHAC/PPO-Chipload): Neural policy weights $\psi^\star$ are trained strictly offline on the dev split. Held-out test tasks evaluate true out-of-distribution geometric generalization, either zero-shot ($K = 0$ simulator backward passes on test) or via controlled inference-time trajectory overfitting ($K > 0$ steps on test, Exp3 & Exp5b).
- Automated Multi-Tool TAMP (Exp6): Requires zero training split; given an unseen .stl mesh and alloy, feature recognition, tool-library selection, and persistent-IPW optimization execute directly on the input part.

C. Physical Motivation and Definitions of Evaluation Metrics

Every reported metric corresponds to a specific shop-floor productivity, surface-integrity, or machine-safety criterion:

- Blended Cycle Time T_{cycle} (s) and Gain $-\Delta T$ (%): Total program execution time under acceleration-bounded corner blending (Eq. (5)), decomposed into Engaged G1 Time ($\Delta V_k > 0$, productive metal cutting), Air-Cut G1 Time ($\Delta V_k = 0$, unproductive feed-rate traversal through previously cleared space), and Rapid G0 / ATC Time (non-cutting repositioning and 6.0 s tool swaps).
- Fair Same-Speed Time T_{same} vs. Optimal-Chipload Time T_{opt} (s): Because naive nominal CAM uses a conservative constant feedrate, cycle-time gains can conflate faster speeds with shorter paths. T_{same} locks all methods' cutting moves to the exact same feeds_and_speeds baseline feedrate f_{base} (11.7 mm/s for ∅5 mm in 6061-T6), isolating pure geometric smoothing and air-shortcut savings; T_{opt} evaluates full engagement-adaptive feed scheduling.
- Target Gouge V_{gouge} and Uncut Residual V_{res} (mm^3): Volume erroneously removed inside the finished CAD part (V_{gouge}, where any value above tolerance scraps the part) and excess stock left unremoved inside the operation target (V_{res}, which stalls downstream finishing tools), audited at 0.25 mm voxel / 0.125 mm step resolution alongside binary holder, rapid-stock, and vise-jaw collision screens.
- Peak MRR (mm^3/s) and Overload (%): Maximum instantaneous volumetric removal rate $\max_t(\Delta V_t/\tau_t)$ and percentage excess above rated $\text{MRR}_{\max}$; exceeding $\text{MRR}_{\max}$ packs flutes with chips and stalls the spindle.
- Actual Undeformed Chip Thickness h_{ex} (mm), RMSE, and Optimal Chipload Band (%): Instantaneous radial-thinning-corrected chip thickness (Eq. (6)), its time-weighted root-mean-square error

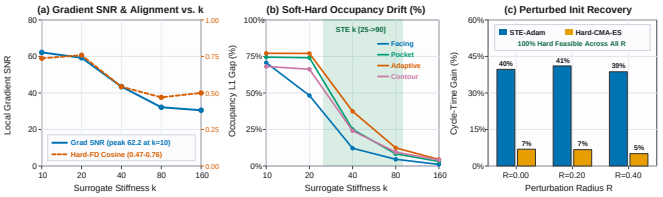


Fig. 2. Exp1: Surrogate gradient conditioning and perturbed recovery. (a) Analytical gradient SNR (62.2 → 30.5, blue) and hard finite-difference cosine (0.76 → 0.47, orange). (b) Soft-to-hard occupancy L_1 drift collapses from 72.6% ($k = 10$) to 8.7% ($k = 80$). (c) Perturbed initialization recovery ($R \in \{0.0, 0.2, 0.4\}$, 48 calls): STE-Continuation-Adam achieves 38.7%–41.1% gains vs. 5.1%–6.9% for Hard-CMA-ES.

against target h_{opt} (0.0291 mm/tooth for ∅5 mm carbide in 6061-T6), and the percentage of engaged cutting time spent inside the healthy $[0.56\,h_{\text{opt}},\ 1.26\,h_{\text{opt}}]$ band (avoiding both rubbing work-hardening below $h_{\min}$ and edge fracture above $h_{\max}$).

- Surrogate Conditioning & Watertight STL ΔV (%): Analytical gradient signal-to-noise ratio (SNR $= \|\mathbb{E}[\mathbf{g}]\|_2 / \sqrt{\mathbb{E}\|\mathbf{g} - \mathbb{E}[\mathbf{g}]\|_2^2}$), cosine similarity to central hard finite differences, soft-to-hard L_1 occupancy drift, and enclosed volume discrepancy ΔV of closed manifold .stl meshes reconstructed independently by CAMotics v1.2.0.

VI. Experimental Results and Analysis

A. Exp1: Surrogate Conditioning and Perturbed Reachability

Motivation: Smooth CSG relaxations face a fundamental trade-off between wide-basin gradient support at low stiffness k and exact Boolean fidelity at high k. Figure 2 quantifies this trade-off across $k \in \{10, 20, 40, 80, 160\}$ on the four dev families. Analytical gradient SNR and directional cosine against hard finite differences peak at $k \in \{10, 20\}$ (SNR = **62.19**–**59.24**, $\cos\theta$ = **0.737**–**0.758**) and remain well-conditioned through $k = 80$–160 (SNR = 32.14–30.53, $\cos\theta$ = 0.47–0.50). Meanwhile, soft-to-hard occupancy L_1 drift collapses from **72.6**% ($k = 10$) to **8.7**% ($k = 80$) and **3.2**% ($k = 160$). Sweeping $k : 25 \to 90$ under STE recovers 100% feasibility and 38.7%–41.1% cycle-time reductions from uniformly perturbed initializations $\|\boldsymbol{\theta}_0\|_\infty \le R \in \{0.0, 0.2, 0.4\}$ (5.8×–7.6× Hard-CMA-ES).

B. Exp2: 16-Task Trajectory vs. RL Benchmark and Physical Ablations

Motivation: Under a strict simulator-invocation budget ($B = 64$ calls, 3 seeds), how do first-order smooth-to-hard trajectory and RL optimizers compare against zero-order RL and evolutionary search operating on exact hard Boolean cutting? Table I and Fig. 3 show that STE-Continuation-Adam ($-\mathbf{42.0} \pm \mathbf{13.5}$%, 73.91 s) and Annealed-Adam ($-\mathbf{43.0} \pm \mathbf{15.7}$%, 73.74 s) outperform fixed-k Naive-Smooth-Adam (−31.9%) by > 10 points and achieve nearly 4× the speedup of Hard-CMA-ES (−11.3%) and Hard-CEM (−9.3%), all at 100% hard feasibility. Under identical 64–64 MLP policies and 64 simulator calls, first-order DiffPolicy-SHAC ($-\mathbf{37.0} \pm \mathbf{12.7}$%) doubles (2.0×) the speedup of zero-order Hard-PPO ($-\mathbf{18.5} \pm \mathbf{6.6}$%) and leads all methods on enclosed Pocket roughing (−**32.1**%) via unsmoothed per-segment feed modulation.

In our 5-arm physical ablation on the 8 test tasks (Table II bottom, Fig. 4b), freezing feeds (Path-Only-Frozen-Feed) without air-detour shortcutting yields only 0.2% gain, joint XY-and-feed STE reduces Peak MRR overload by 29% vs. frozen-path feed scheduling (5.2% vs. 7.3%), and removing the MRR barrier ($w_{\text{MRR}} = 0$) causes catastrophic +**136.7**% cutter overload (262.6 mm^3/s). In Exp3 (Table II top, Fig. 4a), multi-task policies trained on the 8 dev tasks show that Hard-PPO-ZeroShot suffers +24.5% Peak MRR overload on unseen test geometries, whereas 16 analytical gradient calls cut SHAC overload 4× (16.9% → 4.4%).

C. Exp5 & Exp5b: Fair Same-Speed vs. Optimal Chipload and Fast Inference-Time Trajectory Overfitting

Motivation: First, to ensure a strictly fair baseline where speedups are not an artifact of an overly slow nominal feedrate, Exp5 re-evaluates all 8 held-out test parts against a physics-calibrated Nominal-CAM (@2) baseline set to the exact feeds_and_speeds optimal rate (f_{base} = 11.69 mm/s, h_{opt} = 0.0291 mm/tooth), reporting both Fair Same-Speed (T_{same}) and Optimal-Chipload (T_{opt}) metrics (Table III top, Fig. 5a). Second, Exp5b directly tests whether we can combine the < 2.5 ms speed of an offline-trained RL policy with rapid test-time trajectory overfitting on the unseen inference part—and compares this against test-time RL fine-tuning (Table III bottom, Fig. 5b).

Table III (top) reveals three critical physical insights:

1) Why 3D MRR Alone Fails Under Radial Chip Thinning: When re-evaluated under Martellotti chip-thinning mechanics, 3D STE-Adam (MRR) drops to 41.1% optimal-band compliance (h_{ex} RMSE 0.0163 mm, 82.12 s) because capping volumetric MRR without accounting for $\psi(a_e/D) < 1$ underfeeds light radial passes into the rubbing regime.
2) Pure Geometric Gain at Fair Matched Speed: Locking every cutting segment to the exact same feedrate f_{base} across methods, DiffCAM-Chipload still reduces mean cycle time from 55.21 s → **53.60** s (−**2.8**% mean across all 8 parts, and up to −**7.9**% on trochoidal adapt-test-01) purely by smoothing corner junctions, shifting waypoints up to 2.98 mm within safe wall clearance, and collapsing stock-verified empty-air G1 detours into safe G0 bridges.
3) Joint Optimal-Chipload Advantage: Enabling per-station chipload optimization drives DiffCAM-Chipload to **48.07** s (−**14.9**% over physics-

TABLE I

Exp2 Main Benchmark across 16 Milling Tasks (4 Families × 4 Geometries × 3 Seeds, Strict Budget $B = 64$ Simulator Calls). Cycle-time reductions ($-\Delta T_{\text{cycle}}$, %) are mean ± std relative to 10 mm/s Nominal-CAM; all methods achieve 100% hard Boolean feasibility under blended kinematics.

Method	Optimization Class	Facing (%)	Pocket (%)	Adaptive (%)	Contour (%)	Overall $-\Delta T$ (%)	Mean T (s)	Peak MRR (mm^3/s)
Nominal-CAM	Clean-Room CAM Seed	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	122.20	75.4
Hard-CEM [15]	0th-Order Elite Search	15.6 ± 2.6	6.3 ± 3.4	6.1 ± 1.2	9.3 ± 2.0	9.3 ± 4.5	112.17	88.8
Hard-CMA-ES [7]	0th-Order Covariance ES	22.3 ± 2.1	6.3 ± 3.5	6.0 ± 1.8	10.6 ± 3.4	11.3 ± 7.2	110.62	95.9
Hard-PPO [17]	0th-Order Clipped RL	21.0 ± 3.4	20.5 ± 5.6	10.1 ± 4.4	22.2 ± 4.5	18.5 ± 6.6	102.26	93.9
Naive-Smooth-Adam	1st-Order Fixed $k = 25$	33.6 ± 8.1	24.2 ± 14.0	33.3 ± 4.4	36.4 ± 5.2	31.9 ± 9.9	83.44	105.9
DiffPolicy-SHAC [22]	1st-Order Analytic RL	53.6 ± 2.8	$\mathbf{32.1 \pm 12.4}$	24.6 ± 4.2	37.7 ± 2.6	37.0 ± 12.7	81.67	107.4
Annealed-Adam	1st-Order 4:1 k-Sched	57.4 ± 3.1	29.2 ± 19.2	33.0 ± 3.5	$\mathbf{52.4 \pm 3.4}$	$\mathbf{43.0 \pm 15.7}$	**73.74**	109.7
STE-Continuation-Adam	1st-Order 1:1 STE (Ours)	$\mathbf{59.7 \pm 1.2}$	26.9 ± 9.7	$\mathbf{37.3 \pm 7.2}$	44.1 ± 3.7	$\mathbf{42.0 \pm 13.5}$	73.91	113.3

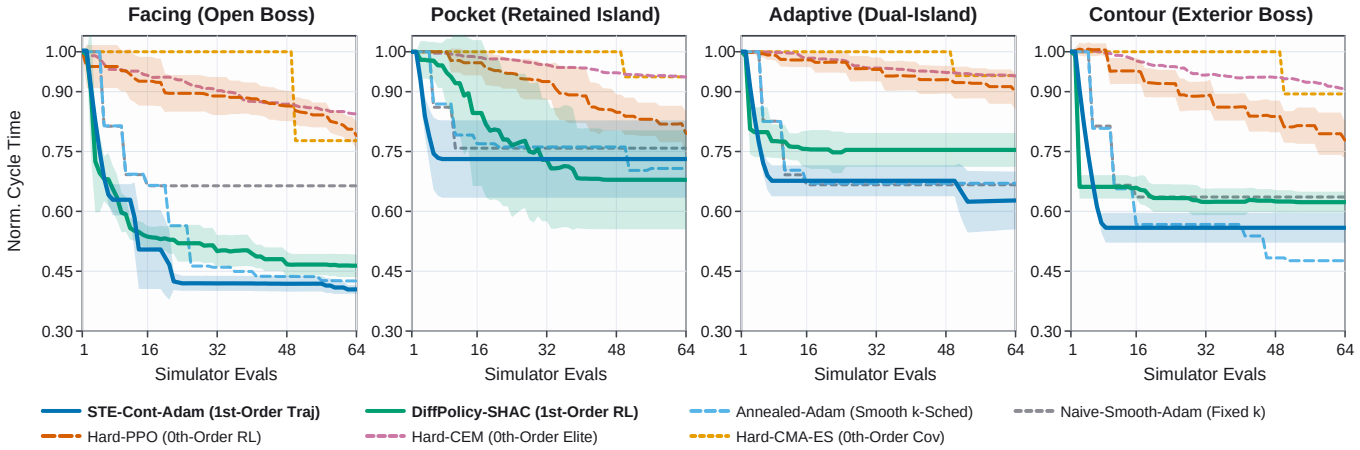


Fig. 3. Exp2 sample-efficiency convergence across clean-room CAM families ($B = 64$ simulator calls, mean ± 1 SD). First-order methods (STE-Continuation-Adam, Annealed-Adam, DiffPolicy-SHAC) achieve steep reductions within 16–24 calls via exact per-segment sensitivities, whereas zero-order RL and ES progress slowly over 100–370 segments.

TABLE II

Top: Exp3 3D Multi-Task Policy Transfer (8 Dev → 8 Unseen Test). Bottom: Exp2 5-Arm Physical & Kinematic Ablations (8 Test, $B = 64$).

Regime / Ablation Arm	Calls	$-\Delta T_{\text{cycle}}$ (%)	Peak MRR	Overload
Exp3: Unseen Test Policy Generalization (3D MRR Surrogate)				
Hard-PPO-ZeroShot	1	26.1 ± 9.5	133.2	24.5%
DiffPolicy-SHAC-ZeroShot	1	8.3 ± 17.9	124.9	16.9%
Hard-PPO-FewShot16	16	23.4 ± 9.5	104.9	2.7%
DiffPolicy-SHAC-FewShot16	16	$\mathbf{27.7 \pm 15.8}$	111.4	4.4%
Scratch-STE-Adam-16	16	$\mathbf{40.1 \pm 7.4}$	114.4	4.0%
Exp2-Abl: Physical & Kinematic Ablations ($B = 64$)				
Path-Only-Frozen-Feed	64	0.2 ± 0.4	82.1	0.0%
Full-Joint-Blended-STE	64	37.0 ± 17.9	114.9	**5.2**%
Exact-Stop-Surrogate	64	39.2 ± 18.2	109.6	2.3%
Feed-Only-Frozen-Path	64	$\mathbf{39.5 \pm 13.7}$	118.7	7.3%
No-MRR-Barrier ($w_{\text{MRR}} = 0$)	64	66.2 ± 3.5	**262.6**	**136.7**%

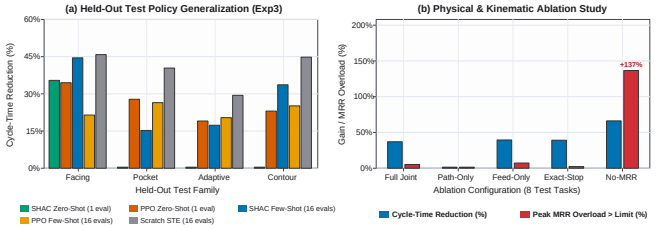


Fig. 4. Exp3 policy generalization and physical ablations. (a) Held-out test cycle-time reductions across families for zero-shot (1 call) and few-shot (16 calls) transfer. (b) Ablating feed optimization collapses gains to 0.2%; ablating the MRR barrier triggers +136.7% cutter overload.

calibrated CAM, 100% posted Haas pass) while keeping **97.7**% of engaged cutting inside the optimal undeformed chip-thickness band (h_{ex} RMSE **0.0053** mm), compared to 42.7% / 42.6% optimal-band compliance for zero-shot SHAC-Chipload and PPO-Chipload.

In Exp5b (Table III bottom, Fig. 5b), we test fast inference-time overfitting on each unseen test part (latencies include one-time ∼ 0.32 s CWE dexel initialization for $K > 0$). Zero-shot SHAC executes its neural forward pass in **3.5** ms, delivering instant speedups on canonical bosses/contours (e.g., 41.87 s → 29.74 s on face-test-00) for an +**8.3**% mean gain, but degrades on unseen long-horizon trochoidal channels (adapt-test-01) and achieves only 42.4% optimal h_{ex} compliance. Attempting to overfit PPO to the test trajectory at inference time via 16–160 simulator rollouts (PPO-Overfit) consumes up to 3.52 s while failing to improve (+8.3% → +7.1%, 41.0% optimal h_{ex}), confirming that high-dimensional ($D \approx$ 250–650) zero-order RL cannot practically adapt per part at planning time. Conversely, warm-starting from SHAC-ZeroShot and running $K = 8$–32 analytical CWE STE-Adam updates directly on $\boldsymbol{\theta}$ (SHAC+Overfit) executes in **0.48**–**0.93** s, repairs out-of-distribution channel tra-

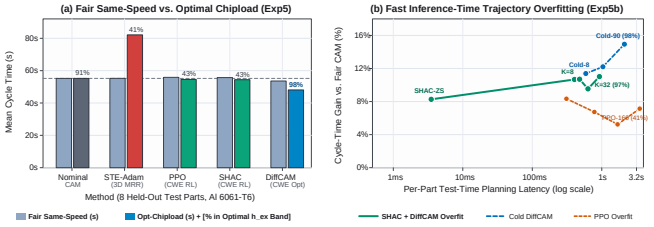


Fig. 5. Exp5 fair chipload comparison and Exp5b inference-time trajectory overfitting. (a) Mean Same-Speed (T_{same}) and Optimal-Chipload (T_{opt}) cycle times across 8 unseen test parts, annotated with % of engaged cutting inside the optimal h_{ex} band. (b) Test-time latency vs. cycle-time gain: warm-starting from SHAC-ZeroShot and running $K \in \{4, 8, 16, 32\}$ analytical CWE trajectory-overfitting steps rapidly dominates zero-order PPO-Overfit (16–160 rollouts) and converges faster than cold-start DiffCAM.

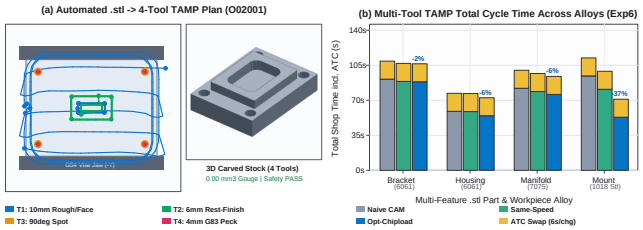


Fig. 6. Exp6 automated multi-tool TAMP demonstration. (a) From a raw binary .stl (tamp-flanged-bracket), DiffCAM recognizes top stock, exterior boss steps, filleted pocket corners, and flange through-holes, selects 4 tools from StandardToolLibrary (T_1 ∅10 mm rough/face/contour, T_2 ∅6 mm IPW rest-finish, T_3 90° spot, T_4 ∅4 mm G83 peck drill), and carves the part with 0.00 mm^3 gouge (O02001). (b) Total shop cycle time (including 6 s ATC swaps) across four parts and three alloys.

TABLE III
Top (Exp5): Fair Same-Speed (T_{same}) vs. Optimal-Chipload (T_{opt}) CWE Comparison across 8 Unseen Test Parts (Al 6061-T6, $f_{\text{base}} = 11.7$ mm/s). Bottom (Exp5b): Fast Inference-Time Trajectory Overfitting vs. Test-Time RL.

Method / Test-Time Regime	Evals	Latency	T_{same} (s)	T_{opt} (s)	$-\Delta T_{\text{opt}}$	Opt h_{ex}
Exp5: Fair Baseline vs. Chipload Optimization (8 Test Parts, 100% Posted Pass)						
Nominal-CAM (@2)	0	—	55.21	55.21	0.0%	90.9%
STE-Adam (3D MRR)	64	1.6 s	55.24	82.12	−38.7%	41.1%
PPO-Chipload (RL)	0†	2.1 ms	55.86	54.57	+9.3%	42.6%
SHAC-Chipload (RL)	0†	3.5 ms	55.71	54.41	+9.5%	42.7%
DiffCAM-Chipload ($K=90$)	90	2.12 s	**53.60**	**48.07**	**+14.9%**	**97.7%**
Exp5b: Fast Test-Time Trajectory Overfitting vs. RL (†trained offline on dev)						
PPO-Overfit ($K=16$)	16	0.78 s	56.02	56.85	+6.7%	40.5%
PPO-Overfit ($K=160$)	160	3.52 s	56.03	56.25	+7.1%	41.0%
SHAC+Overfit ($K=4$)	4	0.40 s	56.70	52.27	+10.7%	62.4%
SHAC+Overfit ($K=8$)	8	0.48 s	56.12	51.29	+10.7%	80.7%
SHAC+Overfit ($K=16$)	16	0.63 s	55.20	51.14	+9.5%	95.6%
SHAC+Overfit ($K=32$)	32	**0.93** s	55.19	50.26	+11.0%	97.0%
Cold-DiffCAM ($K=8$)	8	**0.59** s	55.15	50.16	+11.4%	98.2%
Cold-DiffCAM ($K=32$)	32	**1.04** s	**54.45**	**49.58**	**+12.2%**	**98.5%**

jectories (55.10 s → **50.26** s, **+11.0%**, cutting cross-part gain SD from 23.5% to 9.4%), and lifts optimal-chipload compliance from 42.4% to **95.6%**–**97.0%**. Even more strikingly, starting directly from the unbiased clean-room CAM seed (Cold-DiffCAM) reaches **50.16** s (**+11.4%**, **98.2%** optimal h_{ex}) in just $K = 8$ steps (**0.59** s) and **49.58** s (**+12.2%**) at $K = 32$ (**1.04** s), converging to **48.07** s (**+14.9%**) at $K = 90$ (2.12 s).

D. Exp4: Multi-Controller NC and CAMotics Watertight STL Audit

Table IV verifies all 8 held-out test parts compiled across 6 industrial CNC dialects (RS274, Haas NGC O01001–O01008, FANUC, Robodrill, Makino, Mazak; 48 programs). Every program passes our 0.25 mm-voxel, 0.125 mm-step posted_evaluator with **0.00** mm^3 gouge and **0.00** mm^3 residual, and containerized CAMotics v1.2.0 reconstructs **8/8** closed watertight STL solids matching nominal enclosed volume to $\Delta V = \mathbf{0.000}\%$.

E. Exp6: Automated Multi-Tool .stl-to-G-Code TAMP Demonstration

Motivation: Finally, Exp6 evaluates our overarching thesis from Section I: can a user drop a complete multi-feature 3D part (.stl) and alloy specification into DiffCAM and automatically receive a collision-free, multi-tool CNC program without manual tool selection, feature association, or feed/speed lookup?

Figure 6 and Table V report end-to-end automated execution across four multi-feature parts and three alloys (Al 6061-T6, Al 7075-T6, and Mild Steel 1018). Without human intervention, cam.tamp_planner sizes the billet and parallel vise jaws, recognizes all planar, boss, pocket/island, and hole features (9.4 s mean end-to-end planning + optimization wall time), and adapts both discrete tool selection and continuous feeds_and_speeds physics to each geometry and material:

- On tamp-flanged-bracket, tamp-manifold-plate (7075-T6), and tamp-stepped-mount (1018 Steel, $r_{\text{min}} = 1.5$ mm corners), TAMP reuses the rigid facing endmill (T_1, ∅10 mm, costing 0 s extra ATC swap) to hog out bulk pocket volume before switching to a smaller rest-finishing endmill (T_2, ∅6 mm or ∅4 mm), 90° spot drill (T_3), and diameter-matched twist drill (T_4, ∅3.5–6.0 mm G83 peck), whereas on the narrow island channels of tamp-sensor-housing it skips dual-tool pocket roughing (4 ops vs. 5–6 ops) and automatically derates spindle speed in 1018 Steel (4024–4713 RPM vs. 8000 RPM in 6061-T6).
- Propagating the persistent 2.5D IPW across operations allows T_2 rest-machining to shortcut directly across pocket interiors already cleared by T_1 (slashing T_2 rest-finishing on tamp-stepped-mount from 53.39 s → 40.19 s at fair matched speed and **13.88** s under optimal chipload), cutting mean total shop time (including 18.0 s of fixed ATC swaps) by **4.4%** (up to **11.9%**) at fair matched speed and **12.8%** (**15.4%** cutting-time reduction; up to **36.7%** total / **43.7%** cutting in 1018 Steel) under optimal chipload (**76.8%** optimal h_{ex} band), with **4/4** safety pass (0 vise, holder, or rapid collisions and **0.00** mm^3 gouge) in Haas NGC programs O02001–O02004.

VII. Limitations and Conclusion

Limitations and Path to Universal Zero-Effort Machining. While DiffCAM demonstrates end-to-end .stl-to-multi-tool G-code automation on single-setup prismatic

TABLE IV
Exp4 Independent High-Resolution Verification, 6-Controller Audit, and Haas Mini Mill NGC Shop Packet Summary Across All 8 Held-Out Test Parts. Audited across 6 dialects (48/48 pass) via a 0.125 mm-step posted_evaluator (0.25 mm voxels) and containerized CAMotics v1.2.0.

Haas Prog.	Case ID	CAM Family	Stock (mm)	T_{nom} (s)	T_{DiffCAM} (s)	$-\Delta T$ (%)	Gouge (mm^3)	Res. (mm^3)	6-Dialect	CAMotics STL	STL Vol. (mm^3)
O01001	face-test-00	Facing Boss	34×24×8	79.92	32.85	58.9%	0.00	0.00	6/6 Pass	Watertight (74.9k△)	4245.9
O01002	face-test-01	Facing Boss	32×24×8	77.68	30.34	60.9%	0.00	0.00	6/6 Pass	Watertight (72.6k△)	4370.6
O01003	pocket-test-00	Pocket Island	32×28×8	117.24	96.26	17.9%	0.00	0.00	6/6 Pass	Watertight (107.9k△)	5034.5
O01004	pocket-test-01	Pocket Island	34×26×8	89.94	62.01	31.0%	0.00	0.00	6/6 Pass	Watertight (102.1k△)	5514.2
O01005	adapt-test-00	Adaptive Trochoid	38×30×8	188.64	123.36	34.6%	0.00	0.00	6/6 Pass	Watertight (129.3k△)	6675.4
O01006	adapt-test-01	Adaptive Trochoid	36×32×8	280.60	198.18	29.4%	0.00	0.00	6/6 Pass	Watertight (127.1k△)	7060.9
O01007	contour-test-00	Contour Boss	32×26×8	107.81	59.82	44.5%	0.00	0.00	6/6 Pass	Watertight (76.1k△)	4143.0
O01008	contour-test-01	Contour Boss	32×24×8	88.60	54.19	38.8%	0.00	0.00	6/6 Pass	Watertight (73.2k△)	4204.8
Mean / Pass	8 Held-Out Test Parts (36.2% Agg. Time Saved)			**128.80**	**82.13**	**39.5%**	**0.00**	**0.00**	**48/48**	**8/8** Watertight	$\Delta V = 0.000\%$

TABLE V
Exp6 End-to-End Automated Multi-Tool TAMP Pipeline from Raw .stl Input Across Four Parts and Three Alloys. Total times include 6.0 s physical ATC swaps (Tn M6, G43 Hn); all Haas NGC programs (O02001–O02004) pass chronological vise-jaw, holder, rapid, and gouge audits.

Haas Prog.	Part ID (.stl)	Alloy	Auto-Selected Tools / Ops	ATC (s)	T_{naive} (s)	T_{same} (s)	T_{opt} (s)	$-\Delta T_{\text{total}}$ / $-\Delta T_{\text{cut}}$	Opt h_{ex}	Safety / Gouge
O02001	tamp-flanged-bracket	Al 6061-T6	4 T/6 Ops (⌀10, ⌀6, Spot, ⌀4 G83)	18.0	109.11	106.84	106.48	2.4% / 2.9%	75.6%	PASS / 0.00 mm^3
O02002	tamp-sensor-housing	Al 6061-T6	4 T/4 Ops (⌀10, ⌀6, Spot, ⌀5 G83)	18.0	76.95	76.73	72.54	5.7% / 7.5%	88.8%	PASS / 0.00 mm^3
O02003	tamp-manifold-plate	Al 7075-T6	4 T/5 Ops (⌀10, ⌀6, Spot, ⌀3.5 G83)	18.0	100.00	96.76	93.88	6.1% / 7.5%	76.4%	PASS / 0.00 mm^3
O02004	tamp-stepped-mount	Steel 1018	4 T/5 Ops (⌀10, ⌀4, Spot, ⌀6 G83)	18.0	112.32	99.00	**71.07**	**36.7%** / **43.7%**	66.3%	PASS / 0.00 mm^3
Mean / Pass	4 Multi-Tool .stl Parts (646-Tool Catalog)		**4.0** T/**5.0** Ops	**18.0**	**99.59**	**94.83**	**85.99**	**12.8%** / **15.4%**	**76.8%**	**4/4** PASS

parts with pockets, islands, exterior steps, and holes, four open challenges separate our prototype from universal "slicer-like" 3-axis automation: (1) Multi-setup 6-sided flipping and custom soft-jaw synthesis for parts with undercuts or bottom-side features; (2) Simultaneous 3-axis ball-endmill sculptured surfacing with continuous axial z optimization; (3) Closed-loop frequency-domain chatter and thin-wall deflection modeling [1]; and (4) Physical Haas Mini Mill NGC cutting trials measuring spindle load, cycle time, and surface roughness R_a.

Conclusion. We challenged the manual paradigm of 3-axis CNC programming with DiffCAM, unifying automated task-level setup and clean-room multi-tool selection with classical CAM trajectory initialization, smooth-to-hard differentiable cutting simulation, and learning-based co-optimization. Across 6 experiments, DiffCAM delivers 42% mean single-tool speedups, 2.0× RL sample-efficiency gains, 97.7% optimal chipload compliance, millisecond warm-start trajectory overfitting, and automated .stl-to-Haas-NGC multi-tool programs verified in 6-dialect and CAMotics simulation.

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